CollabX – Project Collaboration Platform

1. Introduction

Good morning everyone.

Today, we are going to present our project, **CollabX – Project Collaboration Platform**.

CollabX is a web-based platform designed to connect students, mentors, and companies in a single environment. Students can share their project ideas, find teammates, and collaborate with others. Mentors can guide students, while companies can explore student projects and discover new talent.

The main objective of our project is to encourage teamwork, promote innovation, and bridge the gap between academic projects and industry requirements.

2. Technologies Used

We have divided our project into three main parts: frontend, backend, and database.

A. Frontend Technologies

1. HTML

HTML is used to structure the web pages. It provides the basic elements of the website, such as headings, forms, buttons, input fields, and content sections.

2. CSS

CSS is used to style the website. It helps us design layouts, colors, spacing, buttons, cards, and responsive page elements.

3. JavaScript

JavaScript is used to implement interactive functionality. It handles user actions, form submissions, data processing, and communication with the backend.

4. React.js

React.js is a JavaScript library used to build the user interface. It allows us to divide the interface into reusable components, manage application state, and update the page dynamically without reloading the entire website.

B. Backend Technologies

1. Java

Java is the main programming language used for backend development. It handles application logic, processes requests, and manages communication between the frontend and database.

2. Spring Boot

Spring Boot is a Java framework that simplifies backend development. It helps us create REST APIs, organize application functionality, and handle HTTP requests from the frontend.

3. REST APIs

REST APIs allow the frontend and backend to communicate. For example, when a student submits a project idea, the frontend sends the project details to the backend through an API request. The backend validates the request and processes the information.

4. Authentication and Session Management

Authentication identifies users who log in to the platform. Session management helps the application maintain the user's login state while performing actions such as creating project ideas or managing collaborations.

C. Database Technology

MySQL

We use MySQL as our relational database management system. It stores and manages information such as registered users, project ideas, and collaboration-related records.

The backend communicates with MySQL to insert new records, retrieve information, and update existing data.

3. Frontend Code Explanation

The frontend code is located in the frontend folder.

The main application interface is implemented in src/App.jsx.

This file contains the React components and application logic used to display the platform and handle user interactions.

The frontend is responsible for the following tasks:

- Displaying the dashboard and project listings.
- Providing registration and login forms.
- Allowing users to submit project ideas.
- Displaying project details and collaboration options.
- Handling navigation and button interactions.
- Sending requests to the backend APIs.
- Updating the interface based on the response received from the backend.

React state management helps the interface respond to user actions. For example, when a user searches for a project or opens a form, React updates the relevant part of the interface.

The styling files contain CSS rules that control the appearance of the website.

4. Backend Code Explanation

The backend code is located in the backend folder.

It is developed using Java and Spring Boot.

The backend processes requests received from the frontend and applies the required application logic.

Its main responsibilities include:

- Processing login and registration requests.
- Validating user information.
- Managing authenticated user sessions.
- Receiving project idea submissions.
- Retrieving project information.
- Processing collaboration-related requests.
- Communicating with the database.
- Returning responses to the frontend.

The backend is organized into different Java classes and controllers. Controllers define API endpoints that the frontend can call to perform particular operations.

For example, when a user clicks the Create Project button, the frontend sends the project details to the backend. The backend checks the request, verifies the user's session, processes the project information, and communicates with the database.

Finally, it returns a success or error response to the frontend.

5. Database Code Explanation

The database stores the information required for the platform to operate.

We use MySQL to organize this information into relational tables.

The database can contain records for:

- Registered users and their roles.
- Student, mentor, and company accounts.
- Project ideas and their details.
- Collaboration requests and related information.

SQL is used to create tables and perform database operations such as inserting, retrieving, updating, and deleting records.

For example, when a new user registers, the backend saves the user's information in the database. When a user creates a project idea, the backend inserts the project details into the appropriate table.

When the dashboard needs to display registered users or available projects, the backend retrieves the relevant records from the database.

6. How the Frontend, Backend, and Database Work Together

Our project follows a three-layer architecture.

Layer 1: Frontend

The user interacts with the website through forms, buttons, project cards, and navigation elements.

Layer 2: Backend

The backend receives the request, validates the information, checks authentication when required, and executes the relevant business logic.

Layer 3: Database

The database stores or retrieves the required information and returns the results to the backend.

The backend then sends a response to the frontend, which updates the interface accordingly.

Example: Creating a Project Idea

Let's understand the complete process using the Create Project feature.

1. The student logs in to the CollabX platform.
2. The student opens the project creation form.
3. The student enters the project title, description, technologies, and other required details.
4. After clicking the Submit button, React sends an API request to the Spring Boot backend.
5. The backend checks whether the request contains valid project information and whether the user is authenticated.
6. The backend processes the request and saves the project information in MySQL.
7. The database stores the new project record.
8. The backend returns a response indicating whether the operation was successful.
9. The frontend displays an appropriate message and refreshes the project listing.

This process demonstrates how all three layers work together to provide the project creation functionality.

7. User Roles and Functionality

CollabX supports four main user roles.

1. Student

Students can share project ideas, explore existing projects, and seek teammates for collaboration.

2. Mentor

Mentors can provide guidance and support to students working on projects.

3. Company

Companies can explore project ideas, discover talent, and connect with students.

4. Admin

The administrator manages the platform and oversees its users and activities.

The platform uses user roles to determine which features and actions are available to different users.

8. Advantages of CollabX

Our project offers several advantages:

- It brings students, mentors, and companies together on one platform.
- It makes sharing project ideas easier.
- It encourages teamwork and collaboration.
- It helps students discover people with complementary skills.
- It provides a common environment for academic and industry interaction.
- It organizes project information in a centralized database.

9. Conclusion

To conclude, CollabX is a project collaboration platform developed using React.js, JavaScript, HTML, CSS, Java, Spring Boot, and MySQL.

The frontend provides the user interface, the backend manages application logic and API communication, and the database stores the required information.

By combining these technologies, we aim to provide a platform where students can share ideas, collaborate with teammates, receive mentorship, and connect with industry professionals.

Thank you.